

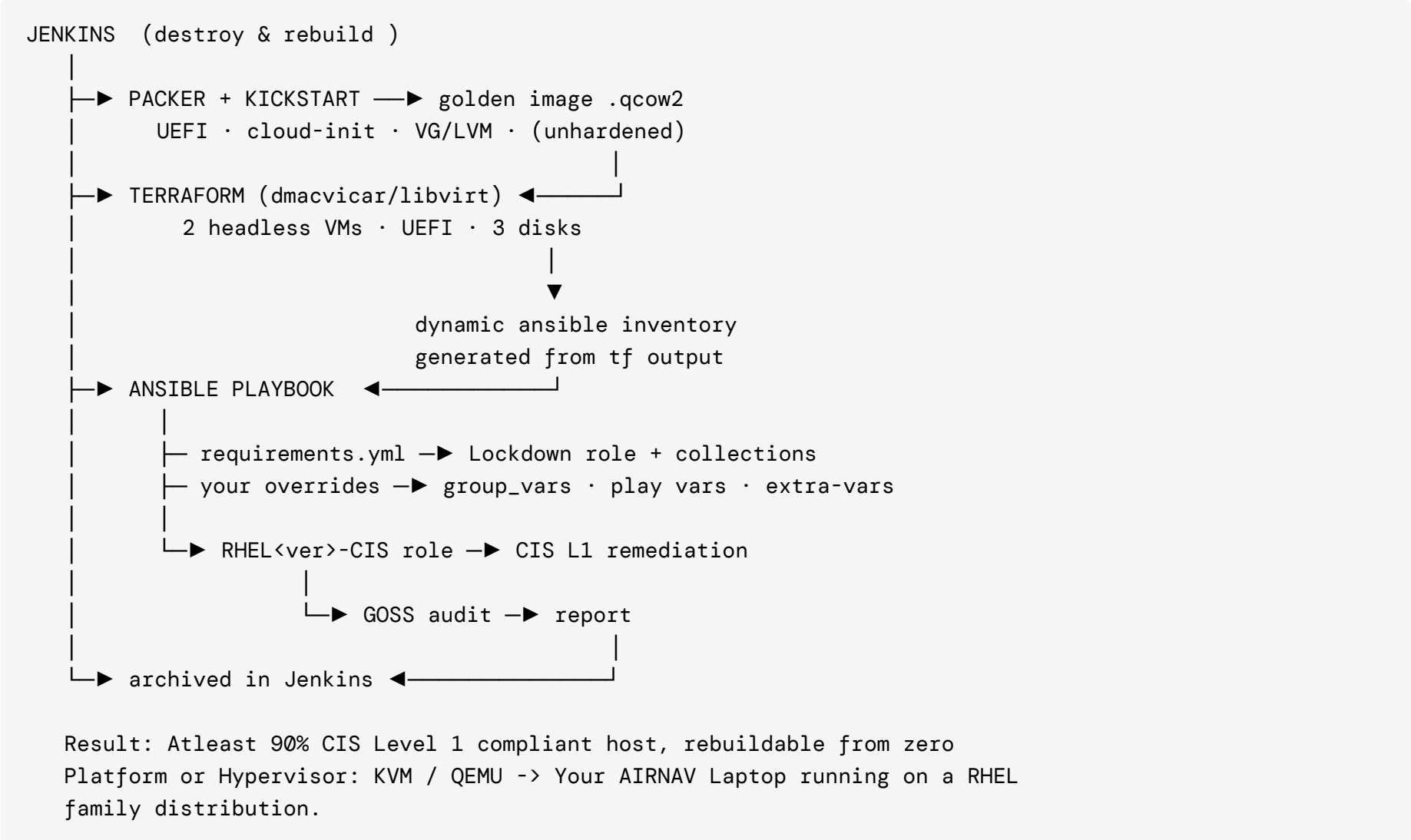
# Trainee Pipeline Task

## Trainee Module: Task Brief

**Duration:** 2 weeks  
**Format:** 2 pairs. Pair deliverable, individual evaluation.  
**Read first:** *Trainee Concepts Guide*

### 1. The build

One pipeline per pair, triggered from a single Jenkins build:



All KVM work is ideally done only via Cockpit and `virsh`.

### 2. Requirements by stage

#### Packer + Kickstart

- Fully unattended AlmaLinux install, boots UEFI
- 20G disk. `/boot/efi`, `/boot`, rest in one VG
- LVs: `/`, `/var`, `/var/log`, `/var/log/audit`, `/var/tmp`, `swap`, plus your assigned extra LV. Root gets a floor plus `--grow`.
- Check which separate-partition rules are actually Level 1 in your role before finalizing the layout. The split differs between RHEL9-CIS and RHEL10-CIS.
- You can add additional configuration into your kickstart file (e.g., provisioning a user with root privileges, packages and dependency installation) as long as justified and well documented

#### Terraform (dmacvicar/libvirt)

- 2 VMs, UEFI (`firmware` and `nvr` set explicitly), 2 vCPU, 2G RAM
- 3 disks each: OS disk cloned from the image, plus 2 data disks of 2G on your assigned bus
- Your own KVM storage pool, no `default` storage pools
- Sets hostname, SSH key, networking
- Terraform output generates the Ansible inventory dynamically.
- `terraform destroy` leaves nothing behind: no domains, volumes, cloud-init ISOs, or generated inventory

#### Ansible + Lockdown

- Format the newly added virtual disks on the Terraform provisioned VM's and format it using your assigned filesystem ( `ext4` / `xfs` )
- Role pulled via `requirements.yml` , pinned to a release tag. Do not fork and edit it.
- **Level 1 - Server only**
- Override your assigned tailoring variables at **three different precedence levels** and document which wins and why
- ansible-vault for anything sensitive. Vault password from Jenkins credentials, never in Git.

Goss

- Built-in audit via `setup_audit` / `run_audit` , runs after remediation, report archived in Jenkins every run
- Target 90% on Level 1. If you land below, explain the gap. If you hit it by switching rules off, say so, that is a legitimate tailoring decision only if it's documented.

Jenkins

| Parameter           | Effect                                                   |
|---------------------|----------------------------------------------------------|
| DESTROY_AND_REBUILD | <code>terraform destroy</code> , then full fresh rebuild |
| REBUILD_IMAGE       | Rebuild the Packer image first, else reuse               |

Think about where your Terraform state lives. A wiped workspace means a destroy button with nothing to destroy.

3. Pair assignments

|                    | Pair A                                                                                                                                                              | Pair B                                                                                                                                                                 |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OS                 | AlmaLinux 9                                                                                                                                                         | AlmaLinux 10                                                                                                                                                           |
| Lockdown role      | RHEL9-CIS                                                                                                                                                           | RHEL10-CIS                                                                                                                                                             |
| Volume group       | <code>vg_sys_a</code>                                                                                                                                               | <code>vg_sys_b</code>                                                                                                                                                  |
| Extra LV           | <code>lv_opt on /opt</code>                                                                                                                                         | <code>lv_srv on /srv</code>                                                                                                                                            |
| Data disk FS / bus | <code>xfs</code> / <code>virtio</code>                                                                                                                              | <code>ext4</code> / <code>scsi</code>                                                                                                                                  |
| Hostname prefix    | <code>pa-node</code>                                                                                                                                                | <code>pb-node</code>                                                                                                                                                   |
| Storage pool       | <code>pool_a</code>                                                                                                                                                 | <code>pool_b</code>                                                                                                                                                    |
| Tailoring vars     | Login banner, sshd timeout<br><br>Use rsyslog for logging<br><br>No account lockouts on failed password attempts on both user and root accounts (Safety / Security) | Password aging, login banner<br><br>Use journald for logging<br><br>No account lockouts on failed password attempts on both user and root accounts (Safety / Security) |

Find the real variable names in your own role's `defaults/main.yml` . They differ between the two roles, which is the point.

4. Split inside the pair

Each person owns a track. Agree who takes which on day one and tell us.

**The catch:** at the presentation you each explain your **partner's** track, not your own. Plan for that from the start. It is the only reliable way to make sure the knowledge actually crossed over, and it's how we get individual signal out of a pair deliverable.

5. Deliverables

One repo per pair:

```
|— packer/          kickstart.cfg, *.pkr.hcl
|— terraform/      main.tf, variables.tf, templates/
|— ansible/        requirements.yml, playbook.yml, group_vars/, inventory/ (See Ansible Docs for
proper Ansible playbook structure)
|— Jenkinsfile
|— .gitignore
|— docs/design.md
```

`docs/design.md` ,

- Partition layout and reasoning
- How the inventory is generated
- Variable precedence table:

| Variable | Value | Set where | Overrides | Why it wins |
|----------|-------|-----------|-----------|-------------|
|----------|-------|-----------|-----------|-------------|

- Expected Goss results, plus any tailoring decisions or exceptions
- Deviations from this brief, with reasoning. Deviations are fine when defensible.

Plus: archived Goss report from the final run, and commit history that shows who did what. Commit under your own name.

## 6. Presentation

- Repo and architecture walkthrough, each covering the partner's track
- Live: pipeline with `DESTROY_AND_REBUILD` ticked, and while it runs, virsh tasks (prove UEFI, list disks and match them to Terraform, attach and detach a disk on a running VM, walk the pool)
- Goss results, three failures explained, precedence table
- Individual Q&A

## 7. Working agreements

- Git from the start, commit under your own name, small commits
- No plaintext secrets. Vault for values, `.gitignore` for keys and password files.
- Use whatever tools help you, including AI. You'll be asked to explain every decision live, so make sure you understand what lands in the repo.
- Ask questions and collaborate.

## 8. Pace

| Week | Target                                                                                                                                    |
|------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Image builds unattended, boots UEFI, correct LVM. Should have a golden image generated already. Terraform provisioning, Dynamic Inventory |
| 2    | Hardening runs end to end. Goss auditing, Jenkins Integration, Presentation                                                               |